

Chain Extender for Flexible Polyurethane Foams

APPLICATIONS

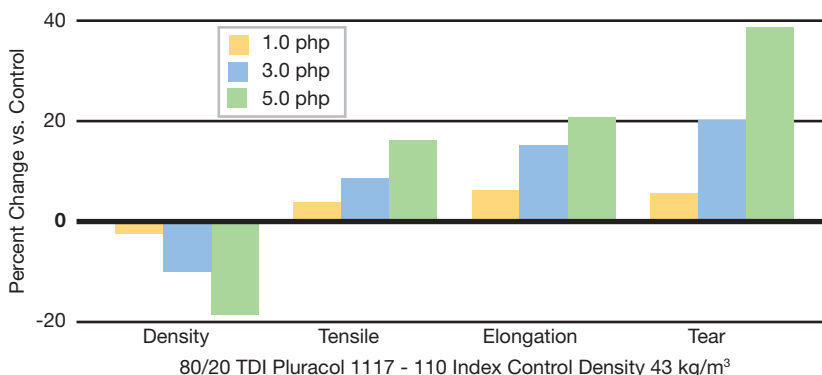
SLABSTOCK FOAM

Unilink® 4200 improves tensile, tear and load-bearing properties in standard TDI foams and high-resilience foam systems, often at lower foam densities and with little effect on reaction kinetics or foam rise. Foams made with Unilink® 4200 have a finer cell structure with less elongation in the direction of rise, which significantly improves compressive strength and dimensional stability. In polyester foams, Unilink® 4200 improves tear strength and load-bearing performance without degrading other foam properties.

COLD MOLDED FOAM

Unilink® 4200 significantly softens cold molded foams and reduces foam density. Finished foams have higher tensile strength, improved tear and elongation as well as better mold flow, and can often be removed from the mold sooner.

Effect of Unilink® 4200 Secondary Diamine on the Properties of HR Slab-stock Foam



Effect of Unilink® 4200 on Flexible Slab-Stock HR Foam Properties Based on Pluracol 1117 Graft Polyether Polyol

Formulation (php)	1	2	3	4
Pluracol 1117	100	100	100	100
Fyrol FR-2	5.00	5.00	5.00	5.00
DC-5043	1.20	1.20	1.20	1.20
Water	2.08	2.08	2.08	2.08
Dabco 33-LV	0.15	0.15	0.15	0.15
Niax A-1	0.05	0.05	0.05	0.05
Dabco T-12	0.40	0.40	0.40	0.40
Unilink 4200	0.00	1.00	3.00	5.00
TDI 80/20	28.9	29.5	30.7	32.0
NCO index	110	110	110	110
Cream time, secs	28	28	26	24
Rise time, secs	110	106	100	91
Tack-free time, secs	525	518	502	480
Density, kg/m³	43	42	40	35
Tensile strength, kPa	172	179	186	200
Elongation, %	225	230	240	245
Tear strength, N/m	367	385	437	507
Compression set, %	11	11	11	11
Humid aged set, %	20	19	17	16
IFD at 25%, kPa	172	179	207	241
IFD at 65%, kPa	434	462	572	690
Sag factor	2.5	2.6	2.8	2.9

HOW TO USE	Typical use levels in flexible foams range from 1 to 5 parts per hundred polyol (php). In TDI foams and polyester foams rates typically range from 3 to 5 php. Rates in cold molded foams typically range from 1-2 php.	
DESCRIPTION	Unilink® 4200 is a unique di-functional aromatic diamine that builds linear polymer networks, allowing the polyurethane formulator to control the degree of cross-linking by adding higher-functioning polyols. It is an easy-to-handle liquid that is low in toxicity (Ames negative) and relatively insensitive to moisture, and it is compatible with a wide range of polyols, co-curatives and other components.	
TYPICAL PROPERTIES	PROPERTY	TYPICAL VALUE
	Appearance	Clear light to dark brown liquid
	Specific gravity @ 26.7 °C (80 °F)	0.98 – 0.99
	Density, g/cm ³ (lb/gal)	0.99 (8.3)
	Flash point, PMCC, °C (°F), min	161 (321.8)
	Water content, ppm, max	500
	Molecular weight	310
	Equivalent weight, min	153
	Hydroxyl number	362
PACKAGING AND SHIPPING	Not regulated for transportation by US DOT	
SAFETY AND HANDLING	Please refer to the current Material Safety Data Sheet for detailed safety, handling and toxicity information.	
TYPICAL SHELF LIFE	1 year	



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